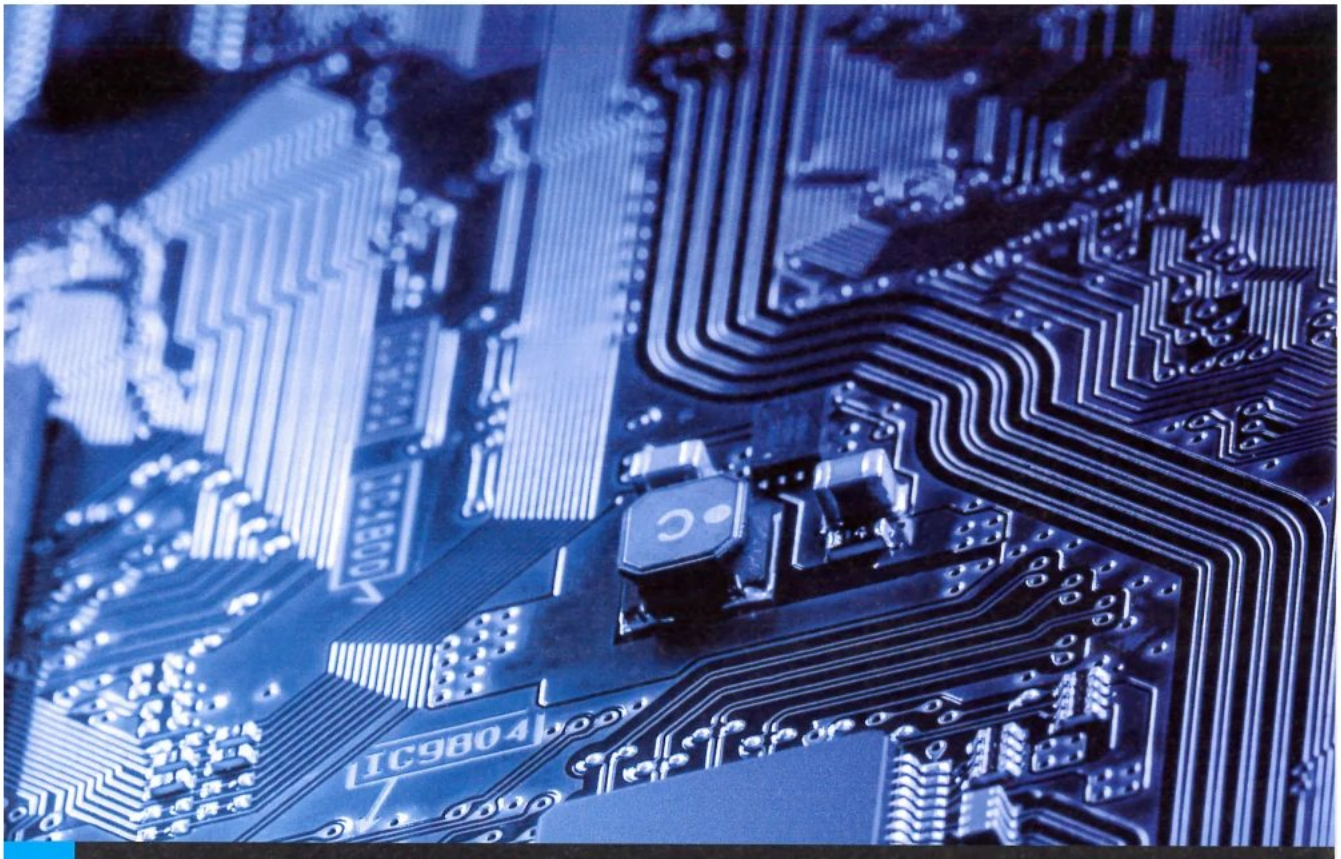


Chapter 5:
Processor fundamentals

Learning objectives

By the end of this chapter you should be able to:

- show understanding of the basic Von Neumann model for a computer system and the stored program concept
- show understanding of the purpose and role of registers, including the difference between general purpose and special purpose registers
- show understanding of the purpose and roles of the Arithmetic and Logic Unit (ALU), Control Unit (CU), system clock and Immediate Access Store (IAS)
- show understanding of how data are transferred between various components of the computer system using the address bus, data bus and control bus
- show understanding of how factors contribute to the performance of the computer system
- understand how different ports provide connection to peripheral devices
- describe the stages of the fetch-execute (F-E) cycle
- show understanding of the purpose of interrupts.



5.01 The von Neumann model of a computer system

John von Neumann was the first person to describe the basic principles of a computer system and its architecture in a publication.

The model von Neumann described has the following basic features.

- There is a processor - the central processing unit (CPU).
- The processor has direct access to memory.
- The memory contains a 'stored program' (which can be replaced by another at any time) and the data required by the program.
- The stored program consists of individual instructions.
- The processor executes instructions sequentially.

5.02 Central processing unit (CPU) architecture

In order to understand how the von Neumann model could be put into practice in a real computer system, we need first to identify the individual hardware components of a CPU and define their functions. Let's consider a system that has the minimum number of components needed. Figure 5.01 gives a simplified schematic diagram of a processor.

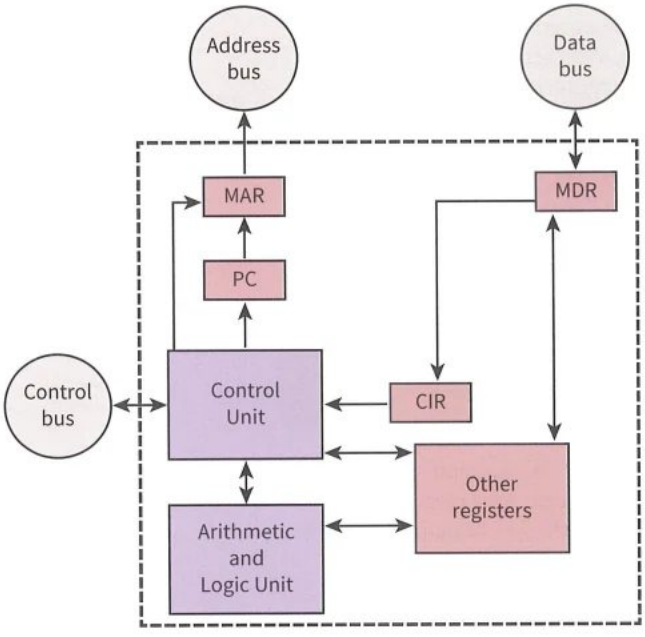


Figure 5.01 A schematic diagram of the architecture of a simple CPU

The dotted outline shows the boundary of the processor. The logical arrangement of some of the processor components is indicated. The arrows show possible directions of flow of data. As the following discussion will show, the data for some of the arrows is actually an address or an instruction. However, in general, data might be an address, an instruction or a value.

The active components of the CPU

The two components of the CPU that have an active role in its operation are the arithmetic and logic unit (ALU) (or Arithmetic Logic Unit) and the control unit. As its name implies, the ALU is responsible for the arithmetic or logic processing requirements of the instructions in a running program. The functions of the control unit are more diverse. One aspect is controlling the flow of data throughout the processor and the rest of the whole computer system. Another is ensuring that program instructions are handled correctly. A vital part of the control unit is a clock that is used by the unit to synchronise processes. Strictly speaking there are two clocks. The first is an **internal clock** that controls the cycles of activity within the processor. The other is the **system clock** that controls activities outside the processor. The CPU will have a defined frequency for its clock cycle, which is usually referred to as the clock speed. The frequency defines the minimum period of time that separates successive activities within the system.



KEY TERMS

Internal clock: controls the cycles of activity within the processor

System clock: controls the cycles of activity outside the processor

Registers

The other components of the CPU are the registers. These are storage components which, because they are placed very close to the ALU, allow very short access times. Each register has limited storage capacity, typically 16, 32 or 64 bits. A register is either general purpose or special purpose. If there is only one general-purpose register it is referred to as the **Accumulator**. Here and in Chapter 6, we assume that the processor has just this one general-purpose register. The Accumulator is used to store a single value at any one time. A value is stored in the Accumulator that is to be used by the ALU for the execution of an instruction. The ALU can then store a different value in the Accumulator after the execution of the instruction.



KEY TERM

Accumulator: a general-purpose register that stores a value before and after the execution of an instruction by the ALU

Figure 5.01 shows some of the special-purpose registers as individual components. The box labelled 'other registers' can be considered to comprise the Accumulator plus the special-purpose registers not identified individually. The full names of the special-purpose registers included in the simple CPU that we are considering are given in Table 5.01 with a brief description of their function.

Register name	Abbreviation	Register's function
Current instruction register	CIR	Stores the current instruction while it is being decoded and executed
Index register	IX	Stores a value; only used for indexed addressing
Memory address register	MAR	Stores the address of a memory location or an I/O component which is about to have a value read from or written to
Memory data register (memory buffer register)	MDR (MBR)	Stores data that has just been read from memory or is just about to be written to memory
Program counter	PC	Stores the address of where the next instruction is to be read from
Status register	SR	Contains bits that are either set or cleared which can be referenced individually

Table 5.01 Registers in a simple CPU

There are three important points to remember. The first is that the MDR must act as a buffer. This is because transfers of data inside the processor take place much more quickly than transfers outside the processor. The second point to note is that the index register (IX) can be abbreviated as IR but in some sources the current instruction register (CIR) is abbreviated as 'IR'. This is a potential cause of confusion. In this book, the index register is always IX and the current instruction register is CIR. Finally, there is also possible confusion if the abbreviation PC is used. This will only be used in this book when register transfer notation is being used, as you will see later in the chapter. Everywhere else, a PC is a computer.

For all of the special-purpose registers, except for the status register, the contents represent one value. For the status register each individual bit is used as a logical flag. The bit is set to 1 if a condition is detected. Examples are the carry flag, the negative flag and the overflow flag.

Chapter 6 (Section 6.07) contains some examples of the use of the accumulator and the status register.

5.03 The system bus

A bus is a parallel transmission component with each separate wire carrying a single bit. It is important not to describe a bus as a storage device. A bus does not hold data. Instead it is a mechanism for data to be transferred from one system component to another.

There will be buses inside the CPU. These are not considered here. The system bus connects the CPU to the memory and to the I/O system. In the simple computer system described in this chapter there will be a system bus that comprises three distinct components: the address bus, the data bus and the control bus. The schematic diagram of the CPU in Figure 5.01 shows the logical connection between each bus and a CPU component. The address bus is connected to the MAR; the data bus to the MDR; and the control bus to the control unit. The system bus allows data flow between the CPU, the memory and input or output (I/O) devices as shown in the schematic diagram in Figure 5.02.